

Developing Views of Nature of Science in an Authentic Context: An Explicit Approach to Bridging the Gap Between Nature of Science and Scientific Inquiry

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ABSTRACT: Reform efforts emphasize teaching science to promote contemporary views of the nature of science (NOS) and scientific inquiry. Within the framework of situated cognition, the assertion is that engagement in inquiry activities similar to those of scientists provides a learning context conducive to developing knowledge about the methods and activities through which science progresses, and, in turn, to developing desired views of NOS. The inclusion of a scientific inquiry context to teach about NOS has intuitive appeal. Yet, whether the learners are students, teachers, or scientists, the empirical research does *not* generally support the claim that engaging in scientific inquiry alone enhances conceptions of NOS. We studied developments in NOS conceptions during a

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science research internship course for preservice secondary science teachers. In addition to the research component, the course included seminars and journal assignments. Interns' NOS views were assessed in a pre/post format using the Views of Nature of Science questionnaire, [VNOS-C] and interviews. Results indicate most interns showed substantial developments in NOS knowledge. Three factors were identified as important for NOS developments during the internship: (1) reflection, (2) context, and (3) perspective. Reflective journal writing and seminars had the greatest impact on NOS views. The science research component provided a context for reflection. The interns' role perspective appeared to impact their abilities to effectively reflect. Interns who assumed a reflective stance were more successful in deepening their NOS conceptions. Those who maintained a scientist's identity were less successful in advancing their NOS views through reflection. In light of these results, we discuss the significance and challenges to teaching about NOS within inquiry contexts. © 2004 Wiley Periodicals, Inc. *Sci Ed* 88:610–645, 2004

INTRODUCTION

The current emphasis on scientific literacy extends beyond calls for knowledge of scientific concepts and methods of scientific investigations. Understanding tenets of scientific inquiry and nature of science (NOS) are at the core of scientific literacy (American Association for the Advancement of Science [AAAS], 1993; National Research Council [NRC], 1996). "NOS" refers to the values and underlying assumptions that are intrinsic to scientific knowledge, including the influences and limitations that result from science as a human endeavor. "Scientific inquiry" refers to characteristics of the scientific enterprise and processes through which scientific knowledge is acquired, including the conventions and ethics involved in the development, acceptance, and utility of scientific knowledge. In a basic form, the intercept of these domains of knowledge (NOS, scientific inquiry, traditional science subject matter), along with an understanding of the utility of that knowledge to the individual and society, represents the conceptual basis for a scientifically literate individual.

Reform efforts around the world stress the importance of developing images of science that are consistent with current scientific practices and constructivist perspectives (AAAS, 1993; Driver et al., 1996; Hodson, 1998; Matthews, 1994; Millar & Osborne, 1998; NRC, 1996; Ryan & Aikenhead, 1992). The National Science Education Standards [NSES] (NRC, 1996) states "students should develop an understanding of what science is, what science is not, what science can and cannot do, and how science contributes to culture" (NRC, 1996, p. 21). Without understanding the values and assumptions of the knowledge and the processes by which the knowledge is created, the learner can do little more than construct an image of science consisting of isolated "facts" void of context that make the knowledge relevant and applicable (Lederman, 1998; Schwab, 1962). As such, reform efforts continue to emphasize a "philosophically valid science curriculum" (Hodson, 1988) to develop learners' epistemological views of science. The importance is clear, but the means to achieve these ends are not.

Students and teachers generally lack adequate understanding of NOS (Lederman, 1992; Pomeroy, 1993; Ryan & Aikenhead, 1992). Naïve NOS views have been attributed, at least in part, to learners' lack of experience conducting scientific investigations (Gallagher, 1991; Schwab, 1962; Welch et al., 1981). Recommendations to improve conceptions of NOS strongly emphasize the use of inquiry in science instruction (AAAS, 1993; NRC, 1996; Schwab, 1962). For teachers to gain the necessary knowledge and skills, inquiry experiences should be provided throughout science teacher education (AAAS, 1993; Gallagher, 1991; NRC, 1996; Schwab, 1962; Welch et al., 1981). Yet, the question remains, "Do learners

necessarily develop NOS conceptions aligned with current perspectives advocated for K-16 learners by engaging in scientific inquiry activities?" The purpose of this study was to examine this assertion in the context of a science research internship course for preservice secondary science teachers.

NATURE OF SCIENCE

A concise description of NOS is often debated among scholars (Loving, 1997; Matthews, 1994) and NOS representations are as dynamic as the knowledge and enterprise of science itself. In the present study, we are not claiming one single "nature of science" to represent all scientific knowledge and disciplines. In describing NOS from epistemological and associated sociological perspectives, Ryan and Aikenhead (1992) include the meaning of science, assumptions, values, conceptual inventions, method, consensus making, and the characteristics of the knowledge produced. General agreement within the current postmodern view acknowledges science as a human endeavor, directed by theory and culture, reliant on empirical observation, and subject to change (see, for example, Casti, 1989; Chalmers, 1982; Kuhn, 1970). These unproblematic, generalized aspects of NOS are relevant to K-16 education (Lederman & Abd-El-Khalick, 1998; Matthews, 1994; Millar & Osborne, 1998; Ryan & Aikenhead, 1992; Smith & Scharmann, 1999), and targeted in this study (Table 1). These aspects should be viewed as interdependent.

SCIENTIFIC INQUIRY

"Scientific inquiry" refers to the methods and activities that lead to the development of scientific knowledge. According to the NSES

[Inquiry] involves making observations; posing questions; examining books and other sources of information to see what is already known; planning investigations; reviewing what is already known in light of experimental evidence; using tools to gather, analyze, and interpret data; proposing answers, explanations, and predictions; and communicating the results. Inquiry requires identification of assumptions, use of critical and logical thinking, and consideration of alternative explanations. (NRC, 1996, p. 23)

Within a classroom, scientific inquiry involves student-centered projects, with students actively engaged in inquiry processes and meaning construction, *with teacher guidance*, to achieve meaningful understanding of scientifically accepted ideas targeted by the curriculum (Krajcik et al., 1994; Minstrell & van Zee, 2000; NRC, 1996; Roth & Roychoudhury, 1993).

Authentic scientific inquiry is that which scientists conduct in everyday practice (Roth, 1995). Descriptions of scientific inquiry provided by contextually based laboratory studies (e.g., Knorr-Cetina, 1999; Latour & Woolgar, 1979) offer a glimpse into the complexities of authentic practice. The real world of science is not typically represented in the classroom (Chinn & Malhotra, 2002; Driver et al., 1996; Roth, 1995; Ryder, Leach, & Driver, 1999). Although both contexts provide opportunities for social construction of understanding, the classroom community rarely promotes complex reasoning and negotiation of meaning as it is expressed within the scientific community (e.g., Chinn & Malhotra, 2002). Nonetheless, scientific inquiry is an important element of the classroom and of science literacy.

The meaning of inquiry in science education can vary as much as the methods of scientific inquiry themselves. Bybee (2000) describes "science as inquiry" as comprising three main elements: (1) skills of scientific inquiry (what students should be able to do), (2) knowledge about scientific inquiry (what students should understand about the nature of scientific

TABLE 1
NOS Aspects and Descriptions That Served as a Basis for Comparison

Aspect	Description
Tentativeness	Scientific knowledge is subject to change with new observations and with the reinterpretations of existing observations. All other aspects of NOS provide rationale for the tentativeness of scientific knowledge.
Empirical basis	Scientific knowledge is based on and/or derived from observations of the natural world.
Subjectivity	Science is influenced and driven by the presently accepted scientific theories and laws. The development of questions, investigations, and interpretations of data are filtered through the lens of current theory. This is an unavoidable subjectivity that allows science to progress and remain consistent, yet also contributes to change in science when previous evidence is examined from the perspective of new knowledge. Personal subjectivity is also unavoidable. Personal values, agendas, and prior experiences dictate what and how scientists conduct their work.
Creativity	Scientific knowledge is created from human imaginations and logical reasoning. This creation is based on observations and inferences of the natural world.
Sociocultural embeddedness	Science is a human endeavor and is influenced by the society and culture in which it is practiced. The values of the culture determine what and how science is conducted, interpreted, accepted, and utilized.
Observation and inference	Science is based on both observation and inference. Observations are gathered through human senses or extensions of those senses. Inferences are interpretations of those observations. Perspectives of current science and the scientist guide both observations and inferences. Multiple perspectives contribute to valid multiple interpretations of observations.
Laws and theories	Theories and laws are different kinds of scientific knowledge. Laws describe relationships, observed or perceived, of phenomena in nature. Theories are inferred explanations for natural phenomena and mechanisms for relationships among natural phenomena. Hypotheses in science may lead to either theories or laws with the accumulation of substantial supporting evidence and acceptance in the scientific community. Theories and laws do not progress into one and another, in the hierarchical sense, for they are distinctly and functionally different types of knowledge.
Interdependence of these aspects	None of these aspects can be considered apart from the others. For example, tentativeness of scientific knowledge stems from the creation of that knowledge through empirical observation and inference. Each of these acts is influenced by the culture and society in which the science is practiced as well as by the theoretical framework and personal subjectivity of the scientist. As new data are considered and existing data reconsidered, inferences (again made within a particular context) may lead to changes in existing scientific knowledge.

inquiry), and (3) a pedagogical approach for teaching science content. The first two elements are clearly articulated in the addendum to the NSES on inquiry (NRC, 2000). The current study focuses on the third element, where the science content is NOS, and the pedagogical approach to teaching this content is scientific inquiry.

INQUIRY AS A CONTEXT FOR LEARNING NOS

From a situated cognition perspective, knowledge is linked to activity and the situation under which the knowledge is acquired (e.g., Brown, Collins, & Duguid, 1989; Lave & Wenger, 1991). Thus, scientific inquiry may provide a viable context for discussion and reflection within which learners can develop NOS conceptions (Carey & Smith, 1993; NRC, 1996; Roth & Lucas, 1997; Ryder, Leach, & Driver, 1999). Different approaches to contextualizing NOS within inquiry have varied in effectiveness.

Review of the Literature on the Use of Inquiry to Teach NOS

Implicit and Explicit Pedagogical Approaches. Some reformed-based curricula assume that through inquiry-based activities alone students develop NOS conceptions aligned with accepted contemporary views. This approach relies on implicit messages within acts of inquiry. An *implicit* inquiry-based pedagogical approach refers to the absence of specific attention to NOS. Certainly, within any science instruction, implicit messages about NOS are communicated. However, the implicit approach assumes that receipt and constructed understanding of NOS is a *natural consequence* of engaging in inquiries. In contrast, an *explicit* inquiry-based pedagogical approach to teaching NOS provides inquiry-based learning opportunities with the added instructional component of specific attention to NOS aspects. The explicit approach advances that improving views of NOS should be planned for through objectives, instructional attention, and assessments. This approach intentionally draws learners' attention to aspects of NOS through discussion, guided reflection, and specific questioning in the context of activities, investigations, and historical examples. The teacher provides learning opportunities, models performance, provides opportunity for practice, assesses student understanding, and provides feedback with reteaching as necessary.

Inquiry-Based Curricula: An Historical Overview. Emphasis on inquiry as pedagogy is not new. According to Yager (1997), early efforts stressing the process skills of scientists were seen in the 1930s. In the wake of the Sputnik success of 1957, the push for improved science achievement and advancement in the United States sparked the development of numerous science curricula focusing on inquiry. Included in the laboratory objectives of the 1960s secondary science curricula, such as BSCS biology and PSSC physics, was the objective of enhancing students' understanding of NOS. Students were to understand "the scientific enterprise, the scientists and how they work, the existence of multiplicity of scientific methods, the interrelationship between science and technology and among the various disciplines of science" (Shulman & Tamir, 1973, p. 1119). Despite ample resources and reputable goals for curriculum development and implementation, these curricula were generally unsuccessful in achieving their NOS learning objectives (Lederman, 1992; Ramsey & Howe, 1969). Their *implicit* NOS pedagogical approaches were ineffective.

Teachers' Experiences with Scientific Inquiry. Failures of inquiry-based programs were attributed to lack of consistent and contemporary philosophy within the curricula (Hodson, 1988) and limitations of teacher knowledge and experiences. "... Modern science

courses have failed to achieve their goals because of inadequacies in the philosophical stance underpinning course design and in the implicit philosophies of science teachers" (Hodson, 1988, p. 35). In addition to knowledge of NOS, a teacher must understand the processes by which scientific knowledge is created to effectively incorporate inquiry-based activities to teach NOS (Gallagher, 1991; Robinson, 1969; Rutherford, 1964). "Prospective teachers have limited knowledge of, and experience with, the processes by which scientific knowledge is generated. This puts serious limitations on their ability to plan and implement lessons that will help the students develop an image of science that goes beyond the familiar 'body of knowledge'" (Gallagher, 1991).

In response, teacher education programs have incorporated inquiry experiences to enhance knowledge of NOS. Teacher education programs that involved explicit instructional attention to NOS in conjunction with the inquiry-based activities were more successful in improving NOS views than programs that provided inquiry experiences alone (Abd-El-Khalick & Lederman, 2000). In support of these findings, Bianchini and Colburn (2000) reported successful outcomes of an inquiry-based science course for liberal arts majors. NOS instruction and guided reflections during several inquiry investigations resulted in more informed views of NOS. The authors suggested that the teacher plays a pivotal role in bringing relevant NOS aspects to students' attention through discussion and questioning. Through explicit/reflective instruction, NOS is viewed as a cognitive learning outcome. Similar outcomes have been reported within other contexts such as history of science courses (Abd-El-Khalick & Lederman, 2000; Matthews, 1994); secondary physics courses utilizing inquiry, history of science, and reflective discourse (Roth & Lucas, 1997); elementary science methods (Akerson, Abd-El-Khalick, & Lederman, 2000); and constructivist-based classrooms (Carey et al., 1989; Carey & Smith, 1993; Smith et al., 2000).

Scientific Inquiry in an Authentic Context. Examination of the popular inquiry curricula, such as the early BSCS and PSSC, revealed limitations of the activities as high-level "scientific inquiry" (Herron, 1971; Shulman & Tamir, 1973). The level of inquiry was more aligned with verification and demonstration activities, with minimal resemblance to authentic activity. More recently, Chinn and Malhotra (2002) described disparate complexities and epistemological representations among school-based and authentic scientific inquiry activities. Given the situational and contextual variances between inquiry as expressed in the school community versus the scientific community, we raise the question, "What does one learn about NOS within an authentic context of scientific inquiry?"

Efforts addressing limitations of school-based inquiry involve students and teachers in science apprenticeships. Lave and Wenger (1991) referred to the role of "legitimate peripheral participation" and movement toward the "core" in one's learning about the scientific community and practice through apprenticeship. Surprisingly few investigations have examined specific NOS learning outcomes from apprenticeship experiences (Barab & Hay, 2001; Bell et al., 2003; Richmond & Kurth, 1999; Ritchie & Rigano, 1996). Barab and Hay (2001) and Ritchie and Rigano (1996) studied middle and high school students, respectively, in research experiences with mentor scientists. Although they suggested these students came to understand aspects of NOS, neither conducted formal NOS assessments. Those studies that have begun to gather empirical data on the impact of authentic inquiry experiences on developments in NOS conceptions offer a glimpse into the effectiveness of this approach. Bell et al. (2003) investigated the impact of an 8-week science apprenticeship program on understandings of NOS and scientific inquiry. High-ability secondary students were exposed to a range of scientific investigations, including development of research methods, data collection, and interpretation. Overall, Bell et al. detected few changes in students' conceptions of NOS and inquiry as revealed by pre- to postapprenticeship assessments. In

general, the students maintained naïve views. No NOS instruction or reflection prompts were provided during this program. In a similar summer science program, Richmond and Kurth (1999) investigated the influence of a research apprenticeship on high school students' understandings of the nature of scientists' work and the students' roles in different communities. Students reflected in journals, during interviews, and group discussions on their views of science. Richmond and Kurth reported gains in students' conceptions of scientific processes, the role of evidence, and the tentative NOS. As part of the apprenticeship model, the authors discuss the importance of students moving from the periphery of the community of science to becoming a part of the community. In doing so, the students reported gaining a sense of themselves as scientists.

Although some authors report gains in NOS learning (Barab & Hay 2001; Richmond & Kurth, 1999; Ritchie & Rigano, 1996), relating outcomes solely to inquiry experiences may be misleading, especially given the contrasting findings of Bell et al. and Richmond and Kurth. Richmond and Kurth provided few details of the journal or discussion activities that accompanied the research component of their program. This leaves open the question of the relative impacts on student learning through the authentic science experiences versus the concurrent explicit instruction. Research on scientists' views of NOS offers further insights. Scientists, those who engage in scientific inquiry in its *most* authentic form, do not necessarily hold contemporary views of NOS (Bell, 2000; Glasson & Bentley, 2000; Pomeroy, 1993).

Do Individuals Necessarily Develop Contemporary Conceptions of NOS by Engaging in Scientific Inquiry Activities? The literature reviewed here strongly suggests that “doing science” is not sufficient in and of itself for developing informed conceptions of NOS. There appears to be a missing, *critical*, factor. It is suggested, and supported empirically, that to be effectively taught, NOS must be viewed as a cognitive learning outcome and addressed explicitly and reflectively within the learning environment (Carey & Smith, 1993; Driver et al., 1996; Lederman et al., 2001). Without explicit attention afforded to relevant aspects of NOS within the context of inquiry experiences, learners' views of NOS likely remain unchanged (Bell et al., 2003; Ryder, Leach, & Driver, 1999).

The Purpose of the Investigation

This investigation examined NOS learning developments and attributions during a science research internship course for preservice secondary science teachers (interns). The course incorporated authentic scientific inquiry experiences, explicit NOS instruction, and guided reflections. The investigation assessed the effects of the course on interns' views of NOS, and identifies influential aspects of the course on learning outcomes. Secondary research questions emerged during the investigation. These include, What are the interns' perceived learning outcomes related to each course component? What personal and/or contextual factors may lead to differential learning outcomes regarding NOS? What effect, if any, did the internship course have on views of scientific inquiry? What effect, if any, did the internship course have on views of teaching NOS?

METHOD

The third author of this report designed and taught the internship course. The second author served as a consultant in course design and implementation and with the research analysis and interpretation. The first author served as the primary researcher, participant observer (PO) in the course, and took primary responsibility for data collection, analysis, and interpretation.

Participants

Participants were 13 secondary preservice science teachers enrolled in a fifth-year, Master of Arts in Teaching (MAT) teacher preparation program in a mid-sized, Western university. All participants had earned a BS degree in a science area, with two also having MS degrees and two with PhD degrees in a science area (Table 2). Biographical and background information was collected during interviews and from enrollment forms. Information on prior science research experiences was elicited through a written assignment and interviews.

TABLE 2
Interns' Biographical Data and Research Experiences

Name ^a (Degree Level)	Past Research Experience	Internship Research Area: Level of Inquiry; Interest in Research Setting; Level of Interaction with Others in Setting
Adam (BS)	Undergraduate courses	Archeology/anthropology: low inquiry; high interest; moderate interaction
Darren (BS)	Undergraduate research in same lab as internship placement	Botany: low inquiry; moderate interest; high interaction
Jay (BS)	Undergraduate courses	Soil science: low inquiry; high interest; high interaction
Kris (BS)	Studied starfish eating habits	Forestry: low inquiry; high interest; high interaction
Laura (BS)	Project in bird behavior during undergraduate studies	Chemistry: moderate inquiry; high interest; high interaction
Rich (MS)	BS Ecology, MS Entomology; thesis on moth diversity	Plant pathology: low inquiry; high interest; high interaction
Alice (BS)	BS Zoology; wildlife technician; work with owls	Zoology: low inquiry; moderate interest; high interaction
Lisa (BS)	Undergraduate project in chemistry	Horticulture: low level of inquiry; moderate interest; high interaction
Kim (PhD)	PhD candidate plant ecology; field research	Plant pathology: moderate inquiry; moderate interest; high interaction
Tim (BS)	Undergraduate courses	Plant chemistry: moderate level of inquiry; moderate interest; high interaction
George (PhD)	PhD Entomology; 16 years in entomology research	Plant pathology/ bacteriology: moderate inquiry; moderate interest; moderate interaction
Brenda (BS)	Undergraduate research 2 years in organic chem lab	Organic chemistry: high inquiry; moderate interest; low interaction
Hank (MS)	BS Geology; MS Paleontology; 10 years experience in geology	Archeology/anthropology: low inquiry; moderate interest; moderate interaction

^aPseudonym.

Brenda and Tim focused in teaching chemistry. All others' primary teaching area was life science.

Context of the Study: The Science Research Internship

NOS was an underlying theme throughout much of the program. The term before the internship course, the preservice teachers began the MAT program with a course that introduced them to NOS and scientific inquiry through generic and content-specific activities. In this initial course, students actively engaged in activities, demonstrations, and debriefing discussions designed to teach NOS aspects that have been identified as important for K-12 science education (Table 1) (for a description of some of the activities, see Lederman & Abd-El-Khalick, 1998). In addition, numerous inquiry-oriented activities addressed the common misconception that scientific knowledge is the product of a single, algorithmic, and strictly objective "scientific method." Following this course, the research internship course provided additional NOS learning opportunities within an authentic science context. There were three main components of the research internship course: (1) the research setting, (2) journals, and (3) seminars.

The Research Setting

All interns were placed with a practicing scientist at the University. Attempts were made to place interns with a scientist in an area of the intern's teaching interest and educational background. For the 10 weeks of the term, the interns spent an average of 5 h a week in the research setting. The constraints of other program requirements did not allow for additional research time. The scientists were informed that the interns were not to be confined to dishwashing or technician duties, but were to be exposed to a range of research practices in as authentic a manner as possible. The scientists were asked to welcome the intern into his/her research setting; discuss the purposes, background, techniques, and direction of the research agenda; and involve the intern in a project if possible. The interns were asked to observe and interact with researchers in the setting and engage in a project if possible in order to gain understanding about authentic research.

We generated descriptions of the research experiences using interview transcripts, journal entries, and seminar transcripts. Interns' activities varied with research settings (Table 2). Those designated as "low inquiry" typically engaged in some aspects of the scientific efforts, but their actions were determined by protocols and conventions of the research setting. They followed predetermined procedures, as in the case of Jay who helped measure the effects of aggregate soil size on nitrification. Jay performed some of the assay procedures, but they were all directed and supervised. Similarly, Adam participated in the analysis of ancient hair samples in an archeology research setting. We consider this experience "low inquiry" because he did not make directional or procedural decisions, yet he interacted with the laboratory personnel and prepared given hair samples for analysis according to set guidelines. Alice had more freedom, but was not knowledgeable enough in the area to conduct her own investigation. She found her time best spent shadowing the researchers, asking questions, and listening to their discussions. Interns in "low inquiry" internships were not responsible for critical decision making. Their context was authentic but quite peripheral.

In contrast, a high inquiry experience included designing and conducting an investigation or a part of an ongoing investigation. Such an experience allows for personal decision making based upon the intern's knowledge and expectations. Only one intern had such an experience. Brenda embarked on her own project in an organic chemistry laboratory.

Her project was based on what the scientist mentor requested, but she conducted a literature review and relevant investigation. She was basically left alone to pursue her project, as *she* wanted. As such, the level of interaction with others in the setting was minimal. Given the time constraints on the internship, and diverse content and research areas represented by the scientist mentors, we did not expect any of the interns to have high inquiry experiences. Brenda's experience is unique and offers an interesting contrast to the others.

Journals

Journals consisted of a research section and a reflection section. The research section contained details of experiences within the research setting including procedures, notes, plans, raw data, and interpretations. The intent was for interns to keep detailed records much like a practicing scientist keeps on a daily basis. Interns responded to sets of focus questions (Appendix A) in the reflection section. The purpose of the focus questions was to facilitate the interns' abilities to make connections between the research experiences and aspects of NOS. The instructor collected the journals periodically throughout the term to provide feedback. Journal questions also served to drive seminar discussions. Given that the interns had already considered the questions in relation to their own settings, they were prepared to discuss the topics and examine their knowledge of NOS as a function of their setting in comparison to other settings.

Seminars

Five 2-h seminars (Appendix B) provided opportunity for interns to discuss research settings and experiences, reflect on their experiences in relation to their views of NOS, and develop connections between their research experiences and science teaching. This was often the only time interns had to talk with each other about their research experiences. Peer interactions consisted of interesting findings from research, open talk of their experiences and reactions, and sharing of journal responses. The instructor had minimal involvement in these discussions, only interjecting when needed to focus interns' attentions and comparisons. For example, in the first seminar, interns described their settings and the types of questions that were being investigated. After everyone shared, interns divided into groups of three and compared group members' research questions by considering: *Are all the research questions of the same type? How are they similar/different? Will they require the same type of investigative approach? Why or why not?* Interns were told,

Begin thinking about what type of research is conducted and what you will be doing in terms of the *type* of science. Recall we talked about before [in the NOS and inquiry course] experimental studies and descriptive studies as examples of different types of investigations. Is there a clear demarcation within your setting? Within any one project? Is there a blend? It is early yet, so you will have to revisit this idea during the term. For now, think about the questions addressed in your setting and how they compare with your group members'.

Seminars also involved sharing reactions to presentations and readings, relative to what the interns could draw upon to support their ideas about NOS. Interns again took main responsibility for the dynamics of the discussions with questions and comments directed toward each other and the task. The instructor and PO guided with questions or comments. The following is an interaction wherein it became apparent that NOS aspects were being forcibly separated, possibly as a consequence of the NOS assignment itself. A student's

detection of his struggles, and question toward his peer, served as a vehicle for explicit guidance. The excerpt is taken from Seminar Four, during the NOS presentations.

[Adam explains how he prepared ancient animal hair samples for identification and cataloguing. Hair samples were assumed to be 10,000–12,000 years old, based on the soil strata. Identification was based primarily on physical similarity with known animal hairs and samples from similar time periods in other regions. His assignment for this presentation was to explain how this research related to science as tentative. He focused on errors related to sterility of the procedures.]

Darren: I had trouble seeing when you addressed tentativeness.

Adam: These are conclusions for what type of animals they are. Like let's say this was a buffalo [holds up a photo of a prepped sample]. The only way to do that is to look at other people's conclusions that theirs was a buffalo. So they are relying on the means available now. If somebody else ten years from now has a better way of cataloguing or a better way of getting smaller amounts of DNA . . .

Darren: I guess maybe I'm confused about what tentativeness is then.

Instructor: Darren has a good question.

Darren: That sounds like inference or something. I thought tentativeness would be more like changes over time like if someone found a better method of looking at the hairs, then what they thought they were would be different.

PO: Tentativeness can be related back to the other aspects as reasons why science is tentative, because it is based on inference; it is subjective; it is empirical. Lets say you have additional information, more empirical data, or you are looking at the same in a different way, with a different perspective. Those changes in the empirical basis, in subjective basis, can lead to changes in conclusions. These tie into why science is tentative. The identification of the hair samples in this lab is based on inference. That inference is grounded in empirical data gathered from multiple sources, no doubt. That all ties into how this type of science is tentative.

Darren: I didn't see how these things could overlap like that. That makes sense now.

Instructor: Is source of error the same thing as tentativeness?

Adam: Error might have something to do with it, but is how they decided to do it. They made assumptions.

Darren: Like they found the pig hair and assumed it was a mistake.

Adam: Yeah, they assumed that the people of 10,000 years ago did not have pigs and so they figured they made a mistake in the analysis and identification.

Later in the same seminar, interns explored their contexts in relation to the empirical NOS. The interdependence of NOS aspects was again apparent and deserving of explicit attention:

Kris: In my lab, one of the operating theories is that increased species diversity and abundance is showing a superior habitat. [She explains how in the study of six logging methods, the rodents who die in the trap and release cataloguing program are used to study effects of logging methods on chipmunk populations.] We species them out and measure everything we can possibly measure about these animals, just in case they ever want any of this data. We can tell you the average tail length of a chipmunk species in 1997. This is important because there are all these habitat models. With ecology, models are the key to everything. Hopefully these models reflect some kind of cause and effect relationship. The problem

comes into these habitat studies is that scientists are the ones who select the variables. So already when we make these models . . . *sorry I am bringing subjectivity in here* . . . we are already deciding what is important and predeciding what is kind of related in these models. There is a lot of inference in habitat studies. The data that I collect can be tied back into several things, like the way the logs are cut. There is a lot of inference, a lot of subjectivity. The scientists work together, but rarely do they see the same things on the charts that are generated from the data in these habitat studies. They argue. It is really interesting.

Alice: Do they assume that all chipmunk species die in the traps equally?

Kris: Yeah. They do. There seem to be more males than females. I don't know if they are more adventurous or more dumb or what. But they do assume that the species get trapped equally. That could be one of the errors.

PO: Just a general question. What do you think empirical basis means?

Kim: The evidence to support what they claim.

PO: OK, so we've got evidence, and you've explained for your setting what that means. I want to go back to what Darren raised earlier about the overlap of NOS aspects. What influences that evidence?

Kris: In order to get the evidence, there has to be an investigation. That design is subjective and based on what has been done before. Like what Adam was saying if the guy before them messed up then their work would be messed up because they base their work on previously accepted conclusions.

PO: Ok, so now the knowledge that is created is based on this evidence that is collected from a certain accepted basis from prior work, that too was based on assumptions. The new investigations cannot be separated from the theories they are based upon. The inferences made to make sense of the data are based on that knowledge also. The types of evidence are important, the assumptions are important; it is hard to separate these aspects. Keep that in mind when thinking about this. Don't force yourself to separate them.

Instructor: When you were talking about the models, any of your data is so dependent on the other ecology studies. It all has to hang together. They are inseparable.

This course concentrated on experiences within the scientific community. The journals and seminars intended to facilitate reflection on these experiences and NOS. Questions posed by the instructor and PO intended to generate discussion and bring forth possible examples and comparisons for interns to consider in efforts to make NOS explicit within the experience.

DATA COLLECTION

Data sources include questionnaires, interviews, journal entries, and participant observations. NOS views were assessed in a pre/post format with the Views of Nature of Science Questionnaire (VNOS-C) and follow-up interviews (Lederman et al., 2002). Journal entries and seminar assignments were photocopied. Seminars were video-recorded and transcribed.

Views of Nature of Science Questionnaire (VNOS-C)

The VNOS-C consists of 10 open-ended questions designed to probe views of specific NOS aspects (Table 1). The questionnaire had previously been validated for use with the intended participants (Lederman et al., 2002). The open-ended nature of the VNOS-C allows

respondents to use their own words and examples, without being forced into a choice and/or words being chosen for them.

Semistructured Interviews

Semistructured interviews were conducted following both administrations of the VNOS-C. During the first interview, the interns were provided their preinternship VNOS-C responses. They were asked to read, explain, and elaborate their responses to each item. In addition, interns were asked if their views were different from what they had represented on the questionnaire. Through such questioning, we aimed to identify any misinterpretations of questionnaire items and clarified ambiguous responses, thus facilitating the generation of faithful representations of interns' understandings of NOS.

The purpose of the postinternship interview was to elicit details of interns' (1) research experiences, (2) perceived learning outcomes and attributions for learning outcomes, and (3) changes, if any, and sources of change in views of NOS. The primary researcher conducted this interview at the completion of the internship, after the postadministration of the VNOS-C, and after the analysis of the first questionnaires, interview transcripts, journal entries, and seminar transcripts. Delaying the second interview allowed the primary researcher to analyze the other data without the influence of participants' self-reports of changes and influences of the internship on their NOS views.

The postinternship interview comprised three phases (Appendix C). First, interns described their research experiences. Second, interns described in what ways, if any, they had benefited from the internship and provided specific examples. The qualitative nature of questioning in these two phases allowed the interns to generate their own list of learning outcomes based on their personal experiences. During the first two phases of the interview, the intern, not the interviewer, first indicated influences of the internship on views of NOS. This was an effort to minimize bias from the researcher's previously generated conceptions about each intern's outcomes.

The purpose of the third phase was to assess final NOS views, characterize changes in views, and identify possible sources for changes. Interns examined their pre- and posttest VNOS-C questionnaires and, in a manner similar to the first interview, were asked to elaborate specific posttest responses. Interns described changes they felt had occurred in their views. They were asked to provide examples and describe experiences to which they attributed changes. For discrepancies between verbal and written responses, interns were shown their written summaries and asked to clarify the discrepancy. Final interviews lasted 30–60 min, were audiotaped and transcribed. The primary researcher was not an instructor in the program and, thus, was not likely perceived as an authority figure. As such, the responses regarding course outcomes were considered valid.

Participant Observations

The role of the primary researcher during the seminars was to observe the instructional episodes, record interactions among interns and instructor, and facilitate discussions through questions and comments. Video recordings and observer notes served as data from these seminar sessions.

Course Summary

The assignment given during the final seminar elicited interns' views on what they had gained from the course in general, and what they had learned specifically about NOS and

scientific inquiry. To increase the validity of responses, the interns were informed that the assignment was a type of course evaluation and would not be read until after grades were reported.

DATA ANALYSIS

The primary researcher performed all data analysis and interpretation. The second author reviewed analyses and interpretations, and results were discussed until consensus was reached. Data from each intern were analyzed separately to generate individual profiles. Comparisons were made among interns to describe generalities across the internship. The interns comprise the population for this study. No generalizations are attempted beyond these participants.

Views of NOS

Data related to interns' NOS views include VNOS-C questionnaires and semistructured interviews, journal entries, and seminar transcripts. We generated comparisons between the interns' views and the NOS aspects as described in Table 1. For each intern, the preinternship VNOS-C responses and corresponding interview transcript were reviewed, and a summary profile generated to represent that intern's initial conceptions. Each aspect was scored with a "+" to indicate the intern's agreement that the particular aspect is representative of NOS, a "++" to indicate the intern's abilities to articulate the meaning of the aspect in his/her own words, or a "+++" to indicate the intern's abilities to articulate the meaning of the aspect in his/her own words and provide examples in addition to those discussed in prior class sessions. Additionally, data were examined for reference to connections among aspects. For example, statements referring to the "tentative NOS resulting from the reliance on subjective methods and inferences" would be considered evidence for a connection between tentativeness, subjectivity, and inference. Postinternship NOS profiles were generated via the same process, utilizing the posttest VNOS-C questionnaires and follow-up interviews. Journal entries, seminar transcripts, and course summaries were analyzed in a similar manner and used for purposes of triangulation with the post-VNOS-C profiles and to help temporally and contextually situate changes.

Changes in Views of NOS

We used five phases of analysis to characterize changes in NOS views. The use of multiple methods enhances the validity of the overall profiles generated from subsequent triangulation (Krathwohl, 1998). Because the journals and seminar discussions were part of the explicit/reflective instructional treatment, these data were rich discursive resources for identifying contexts for change.

Changes across time. The preinternship NOS profile, journal, and seminar transcripts were used to initially identify changes in views of one or more NOS aspects. This analysis was completed before the administration of the second interview. Journal entries and seminar transcripts were analyzed in the order in which they were written or recorded during the term. Changes were sought by comparing each of these data to the interns' preinternship NOS profiles, previous journal entries, and previous seminar transcripts. Evidence of changes, drastic and subtle, was sought and categorized. The data were compared and contrasted several times for category generation and refinement.

Changes pre to post. After completion of the journal and seminar analyses, the second interview was conducted. The delay in conducting and analyzing the second interview was an

attempt to minimize researcher bias in the analysis of the journal and seminar data. Knowing the perceived changes in NOS understanding as stated by the interns could influence the interpretation of other data sources. The postinternship VNOS-C and interview transcript were analyzed in the aforementioned manner to generate profiles of each intern's views of NOS at the completion of the internship. Changes in views were sought through direct comparison between the initial and final VNOS-C profiles. Changes were characterized according to the aforementioned methods.

Perceived changes and attributions. The transcripts from the second semistructured interview and final summary were reviewed for references to perceived changes in NOS views and attributions for those changes. Changes were characterized as in the previous two analyses.

Triangulation. The three independent analyses were triangulated to develop a single, in-depth profile of changes and change attributions for each intern. In the event that the intern's perceived changes (the third analysis) were contrary to those identified through the first two analyses, all data were reviewed again for confirmatory and contradictory evidence. If discrepancies could not be satisfactorily resolved, the statement of perceived change and attributions made by the intern in the final interview may be considered the more valid representation of the intern's final status.

Generation of summaries. Postinternship NOS profiles and the change profiles were used to generate summaries for the entire sample. Similarities and differences among interns were sought and summarized to describe the overall effects of the research internship.

RESULTS

The results are presented in three sections. First, we present NOS views for each intern and the changes in NOS views identified as a result of the internship. Second, we report the non-NOS change outcomes that resulted from the internship. Third, we describe the factors for change, including the identified contributions of each component of the internship.

Views of NOS

Preinternship NOS Views. Prior to the internship, all interns had been introduced to the aspects of NOS emphasized in the program. In their preinternship VNOS-C responses and interviews, all articulated, to some degree, the meaning of most of the aspects. The depth of their initial understandings varied considerably from "mimicry" of definitions to elaborate descriptions. Only four interns initially held misconceptions about theories and laws. One of these interns, Adam, held naïve views about tentativeness and creativity as well. His thoughts regarding "proof" in science indicated a "seeing is knowing" view. Adam stated, "Some things in science can be proven true if we can see it." Such a view disregards the role of inference or creativity in understanding phenomena. The four interns who expressed misconceptions about theories and laws acknowledged a difference between the two, but the difference was in the level of absolute proof. Evidence of this naïve view was seen in Kris's statement, "Laws occur over and over. Theories aren't proven." Responses of three interns indicated rather realist views of science. They consistently indicated that change in science was leading toward a final and absolute description of reality. For example, Hank considered change in science to result from new technology that enabled new data and the correction of previous mistakes. This type of statement alone is not necessarily problematic, but when probed further, Hank explained that mistakes would keep being fixed through new observations until they "finally got it right." The existence, or not, of an accessible objective reality was not a strong focus during the program, but it was interesting that 3 of the 13

participants voiced this view at the start of their internship. None of the interns explicated connections among NOS aspects at this time.

Changes in NOS Views. At the completion of the internship, all participants demonstrated abilities to articulate fairly detailed descriptions of all targeted aspects of NOS. Although, the level of detail still ranged from simple affirmation that the aspect applies to science to very elaborate descriptions indicative of deeper understandings. The three analyses that looked for evidence of change resulted in two main change categories that differ in degree of change, and several subcategories that describe the types of changes. A change was considered (1) “major” if there was evidence of an overt switch from a misconception/naïve view initially to a view that is in more agreement with that promoted in the course, or (2) “enhancement” if an intern demonstrated improved understandings over his/her initial profile, thus indicating a shift in the desired direction. Where interns held (+) or greater initial views, an “enhancement” was identified through (1) more elaborate descriptions of aspects such as the articulation in the individual’s own words as opposed to recalling formal definitions, (2) the ability to make connections between scientific inquiry and aspects of NOS through supporting examples from present or past research settings, or examples different from those presented in course work, (3) a demonstrated shift from viewing aspects of NOS as separate components to realizing the interrelationships of the aspects, and (4) a demonstrated shift from viewing NOS as separate from science content to realizing that NOS is a part of science. Table 3 summarizes changes. Note that the order of interns is by relative extent of change in NOS views, from most change to least.

Eleven of the 13 participants (85%) demonstrated major or enhanced views of NOS upon completion of the course. Of these 11, four (Adam, Darren, Jay, Kris) demonstrated *major* improvements in one or more aspect, including their conceptions of scientific theories and laws. Adam demonstrated the most dramatic change with major shifts in his conceptions of tentativeness, creativity, and the distinction between theories and laws. He also elaborated on all aspects beyond simple definitions and provided representative examples from his research setting to support his views. With regard to tentativeness, subjectivity, and the empirical basis of science, Adam used the example of two competing theories in his setting. The theories offer two explanations for how the Americas were initially populated. Adam explained that researchers in his setting gathered evidence that supported one theory over the other. Eventually, he claimed, the more supported theory could become the dominant theory with the accumulation of substantial evidence.

Connecting NOS to Research Experiences. All 11 who demonstrated a positive change had enhanced views of one or more NOS aspects (Note the “E” in Table 3). Most articulated meanings in their own words and/or were able to include supporting examples from inquiry experiences. The following quotes are representative of this type of connection:

We plot points on a graph to get a linear regression for nitrification ratio. A couple of times there have been some outlier points that skewed the regression line equation. We know that our points lie between a certain spot in the standard curve. We decided this outlier was a mistake and threw it out to fix the regression . . . This also strongly illustrates subjectivity in analyzing data and results. [Jay, journal]

In this placement there are a variety of theories we operate with in designing investigations and inferring results. These theories influence decisions made in the research . . . Habitat studies take for granted theories centered around the idea that animal abundance is a good measure of habitat quality, reproductive success is a good measure of habitat quality, animal

TABLE 3
Change in Views of NOS Aspects, Scientific Inquiry, and Teaching of NOS

Name ^a	Tentative	Creative	Subjective	Empirical	Social/Cultural	Theories/Laws	Obs/Inf	Scientific Inquiry ^b	Teaching NOS
Adam	M (++)	M (++)	E, S	E, S, Ex	E, S	M (++)	E (++)	Multiple methods	Insider's view to present to students
Darren	E, C, Ex	Ex	E, Ex	E, Ex	M (+)	M (++)	E, Ex		Science is a human endeavor and should be taught that way
Jay	E, S, C	E, C, Ex, S	E, C, S	E, Ex, S	E, Ex, S, C	M (+)	E, Ex, S, C	More processes involved	Errors can be valuable for learning
Kris	E, Ex, S	E, S, Ex	E, Ex, S	E, S, Ex	E, S, Ex	M, Ex, S	E, Ex, C, S	Greater complexity of design	Explore and design own study is import for context
Laura	E, Ex	E, Ex, C	E, S	E	E, Ex, S	E, Ex	E, Ex, S	Multiple methods	Teachers are link between students and scientific community; teach NOS and real-world examples
Rich	NC (++)	E, Ex, C, S	E, Ex, S, C	E, Ex	E, Ex	E, Ex, C	Ex, S, C	Multiple methods	Teaching multiple methods is important; explicit teaching of NOS is important; inquiry
Alice	E, Ex, C	E, Ex, S	E	E, Ex, C		E, S	E, S	Multiple methods	Thought about NOS more; model for school inquiry and NOS

Continued

TABLE 3
Change in Views of NOS Aspects, Scientific Inquiry, and Teaching of NOS (Continued)

Name ^a	Tentative	Creative	Subjective	Empirical	Social/Cultural	Theories/Laws	Obs/Inf	Scientific Inquiry ^b	Teaching NOS
Lisa	E, Ex	E	E, Ex	E, Ex	E, E	NC (+)	E, Ex, S	Multiple methods	Interest in teaching NOS increased
Kim	E, C, Ex, S	E	E, C, Ex, S	NC (++)	E, S, Ex	NC (++)	E, Ex, C, S	Multiple methods	Interest in teaching NOS increased
Tim	NC (++++)	NC (++)	Ex	E, Ex, C	NC (++++)	E	S	Purposes can differ	Teaching NOS should be explicit; more comfortable with teaching NOS
George	NC (++)	E, Ex	E, Ex	NC (++)	E, Ex	NC (++)	E, Ex		Increased awareness of NOS teaching
Brenda	NC (+-)	NC (+)	NC (+)	Ex	NC (+)	NC (++)	NC (++)		
Hank	~E	NC (+)	NC (+)	NC (++)	NC (++)	NC (++)	NC (++)	Multiple methods	Won't teach "Scientific Method"; must teach NOS explicitly

Note. M: major change from misconception to informed view (resultant level); NC: no change evident (resultant level); E: enhanced (started out adequate, but gained); S: showed connections to science in general; Ex: presented examples from current or prior research settings; C: showed connections among aspects. Final NOS views: + provides a definition or affirmative response; ++ provides a description in own words, examples from class; +++ provides a description in own words and additional supporting examples; - inconsistent statement or inappropriate example given.

^aParticipants are listed in order of most change to least change in level of NOS understanding.
^bDemonstrated changes in conceptions of inquiry.

diversity is a great measure of complex habitat quality and conversely, structural diversity is good for animal diversity. These are all theories from which habitat researchers base their investigation designs and comparisons . . . Inference is made in interpreting meaning of results, and the basis of these theories give results credibility. [Kris, journal]

Establishing Connections Among NOS Aspects. Other enhancements involved establishing connections among aspects of NOS and then to science in a broader sense. Novel connections were often in relation to specific examples from research settings. Jay demonstrated such an enhanced view. He had entered the course without much confidence in his NOS conceptions. He generally had affirmed that the targeted aspects did apply to science but was unable to elaborate through written or verbal responses. Through his journal writings, Jay demonstrated dramatic change in his supportive discourse. In one entry, he explained how tentativeness, observation, inference, and evidence connected to his setting and to science in general.

Where NOS ideas come into this process is that through every step we have changed our thinking and procedures based on what we have observed and why we think that happened. For example, . . . [presents a lengthy example of a change that occurred in his setting] . . . Science has to be able to change its ideas based on obs. and inferences or it could not progress and develop new ideas, procedures, etc. I know that this connection is very shallow, but it is what I have been a part of in my research setting. This is the manner in which I have been able to observe these things Even though I have not observed tentativeness and obs/inf on a large scale or a more in depth manner, I understand the ideas and significance of these things. It is kind of profound in its simplicity. In science we try to answer questions about our world. To do this we do things like experiments and make observations about what happened. Next we look at the observations (data) and make inferences and conclusions about the reasons for what we observed. This can be in relation to a large concept or idea like evolution or it can be in regard to something smaller like in my experiment where we were setting up a procedure. Based on our observations/inferences, we change our ideas or questions—tentativeness-science is always open to change. This is an awesome thing. We know that we are not always right, and if new ideas and evidence come in observations and inferences lead us to change. Science evolves over time to develop new ideas and concepts all based on observations/inferences and supporting evidence. Three simple ideas with very complex results. [Jay, journal]

Others made similar explications, again prompted through journal reflections:

All phases of the scientific process use imagination and creativity. The design state of an investigation takes a lot of research of . . . related studies—these studies of course are based on subjective scientists who designed that particular investigation. Creativity and inference are most commonly used in interpreting the results. Inferences are based on data observations, inferences depend upon an individual's point of view and previous experiences. Inferences in this lab are based on previous habitat study conclusions It takes creative insight to be able to make inferences, and draw all, or any, of the pieces together. [Kris, journal]

. . . It is science . . . The aspects are most applicable in a real research setting, whether it be a small or large project. We analyze parts of inquiry and NOS but really in research they are *integrated* and central to the doing of science. [Alice, journal]

In the second seminar session, Rich reported seeing a new relationship among NOS aspects. In his presentation, he drew a Venn diagram showing creativity, tentativeness, and subjectivity in three overlapping circles. He said to the class,

What I thought was interesting that I didn't discuss with the group was that we kept coming upon the subjectivity and the tentativeness and the creativity. We kept going back and forth and saying that is more tentative and no that is more subjectivity. So we tried to take these vary overlapping ideas and create a system we can make sense out of. For me sometimes it seems like this amorphous, chaotic, kind of goo with these ideas and it seems as though every time we talk about these things they are constantly overlapping, again and again and again. That is my take on it. [Rich, seminar #2]

Others reported similar realizations resulting from the NOS presentations during Seminar Four (Appendix B). Recall the seminar excerpt presented earlier in this report where Darren raised a question about tentativeness. He had not considered how aspects were related before that discussion.

No Changes in NOS Views. Brenda and Hank did not demonstrate changes (Table 3). These two consistently focused on the content of their research settings throughout their journal writings and seminars. It is interesting to note their initial NOS views were not very developed, and they both presented realist views of science. However, Hank held great confidence in his ideas, owing his views to his research background. When asked how he had benefited from the internship, he responded, "Not much. I've done master's degrees and run research projects, you know. I've done those sorts of things, planned research and written papers." His response suggests he viewed the purpose of the internship as a way to gain experience in doing scientific research. Brenda held a similar view evidenced by calling herself a "mini-graduate student who was working on a project."

Non-NOS Change Outcomes

Other categories of change emerged from interns' self reports. Eight of the 13 (62%) reported a better understanding of the multiple methods applied in scientific investigations (Table 3). Where they had been introduced to this idea during a previous course, the examples from their own and others' research settings seemed to make this point more acceptable and relevant to real science. Other outcomes included consideration of scientists' NOS views and views of NOS pedagogy.

Consideration of Scientists' Views of NOS. Interns talked with the scientists about journal focus questions and/or NOS in general. The interns expressed surprise when they realized the scientists either presented contrasting views of NOS or had not thought about NOS before. It became apparent to the interns that the scientists focused on "doing science":

How little thought to the phrases "nature of science" and "scientific inquiry" is given by people who spend their time in research was demonstrated by the guests in the seminars, and in the chemistry dept. with people I talked to. They are concerned with the doing, not the nature of science. [Tim, summary]

I took away from this experience a piece of how some scientists view their own work, their own science. They are pretty much oblivious to the process. All they see is the goal, to find the answer, and the procedure that comes in between. [Rich, summary]

Views of NOS Pedagogy. Recognition of scientists' perspectives may have impacted interns' views of teaching NOS. Even though NOS pedagogy was not a focus of the course,

all but two interns indicated the experience affected their views (Table 3). Most notable is recognizing what we mean by explicit teaching of NOS and what we overlook by relying on students to learn about NOS through simply conducting inquiries.

The internship did force me to spend a lot of time thinking about the NOS and inquiry. Whereas before I simply “did research,” this time I actually thought carefully about the nature of what I was doing. As a result I think I will teach NOS explicitly in the classroom, whereas before, I probably would’ve done it more implicitly. [Kim, post-interview]

It gives me a real understanding as a teacher that, yeah, this stuff isn’t just a natural outflow of teaching science. [Tim, post-interview]

Overall, I came to value the apprenticeship first when the professors at [my research setting] went off about how valuable they thought the experience would be for new teachers. Then it was interesting to see several of the aspects of NOS appear in the lab. What was interesting, however, was when I would talk to Dr. G about my journal questions and he would be completely unaware of these aspects in his lab . . . My experience during this apprenticeship suggests they [the scientists] are not qualified for teaching—especially if you think the NOS is important for students to know about! [Darren, summary]

Factors for Change

The 11 interns who demonstrated advances in their NOS views reported their views to be more formalized and that the course had “made implicit ideas more explicit,” and “brought thoughts about NOS to the forefront.” A focus of this investigation was to identify factors that effected such desired changes. Table 4 summarizes the benefits and learning outcomes as reported by the interns.

Journals. Eleven of the 13 (85%) attributed advancements, first and foremost, to their journal writings. The questions forced the interns to consider their own views and compare them to actual scientific research.

When we first started, and in fact all through the summer term, I’d heard this NOS, but really it was just this thing, but after summer I still didn’t have much idea of what this mystical NOS thing I was supposed to be getting was. I had this list of 7 things that was a part of it, but that was as far as I could go. I think through writing the journals and actually getting concrete examples, I really understood after all that emphasis. I understand these different aspects of NOS. [Lisa, post-interview]

I think the main things that were helpful were the reflections and talking with people in the department that I was working in about NOS type things was really enlightening to me. [Alice, post-interview]

Seminars. Interns reported the presentations and student discussions to be the most beneficial elements of the seminar portion of the course. As previously described, Rich and Darren realized connections among NOS aspects during the seminars. Although not everyone could identify a single “ah-hah!” episode, others reported having similar learning experiences:

This course has helped in developing my understanding of NOS by providing one more forum for me to think, talk, and write about it. The connection of all the parts being related really hit home when we made the 10-min presentations. No matter who was talking or

TABLE 4
Reported Learning Outcomes from Course Components

Course Component					
Research Setting		Seminars		Journals	
Benefits/ Learning Outcomes	Interns (%) ^a	Benefits/ Learning Outcomes	Interns (%) ^a	Benefits/ Learning Outcomes	Interns (%) ^a
Enhanced NOS views	0	Enhanced NOS views	62	Enhanced NOS views	85
Research techniques/ skills	80	Multiple methods and complexity of inquiry	31		
Specific science content	62	Specific science content (guest speakers)	31		
Considered scientists' NOS conceptions	54				
Provided context for reflection	85				
Valuable interactions with researchers	92				

^aPercentage of interns who reported these outcomes from specific course components.

which one of the aspects of the NOS were talking about, there was always reference to one or most of the time more of the other aspects. [Laura, final summary]

I learned more about each aspect of NOS and it became less nebulous. In particular, the presentations we each gave were valuable. The discussions about assumptions that are made in research but are not talked about was particularly helpful for me in opening my eyes to assumptions in science. [Alice, final summary]

The scientists' presentations brought new perspective and examples with which to compare interns' developing views of NOS and scientific inquiry. It was during these presentations when several interns realized the scientists do not necessarily think about NOS. Furthermore, interns considered their own views of scientific inquiry in the context of the speakers' presentations and realized there are a variety of scientific methods. When reflecting on the speakers, Adam stated,

Scientist don't operate under the same "scientific method." Each scientist has their own questions and motivations when investigating. These unique circumstances create unique needs and therefore unique procedures for conducting scientific investigations. [Adam, summary]

Four interns (31%) reported more interest in the content of the scientists' presentations. Hank reported the science content was the *only* benefit from the whole internship.

The seminars and hearing scientists talk about their work was interesting. That was valuable for kind of keeping up with scientific interests . . . We are scientists in this program and we are not doing science anymore. A lot of times, we give up that connection. We are teachers. It was really interesting to hear what is going on now and directions of research and how that relates to what we are doing. That was really valuable. The other parts, the journal writing and internship, were I thought, something I had to do and I didn't look forward to them and didn't think they were useful or valuable. [Hank, post-interview]

The Research Setting. The research setting had the least direct influence, in and of itself, on NOS learning. None of the interns thought their activities within the research setting *directly* impacted their NOS views. Most reported learning techniques and content. Those new to research, or in a setting much different from previous work, valued such novel experiences. What the research setting did provide, however, was an authentic *context* for reflection: a real research setting with which to apply and revise one's knowledge of NOS. This was viewed as the most beneficial aspect of the research portion of the course relative to NOS learning. The context provided fuel for reflections that deepened 11 of the 13 interns' NOS conceptions. Additionally, the context seemed to make the concept of NOS more real and applicable to science and scientific inquiry.

Writing about it [NOS] in a real life setting gave it validity. It wasn't a list of "this is the NOS" where I would say "well, how am I going to teach that." All of a sudden it was what scientists do everyday. They just don't put a label it. When I asked her [the scientist] about tentativeness, she gave me this blank look of "what are you talking about" There was no label on it. NOS is assumed in science investigations. I guess that is what I learned is that I could describe it more than I thought I could by finding examples. [Kris, post-interview]

My understanding of NOS has increased with this course. I don't think it is a result of the research as much as that we were forced to look at and think about these different aspects of NOS . . . I think I did learn more about NOS because of the journals we had to do. They made us apply those concepts to what we were doing in our labs. So for me at least I was able to get a better handle on that stuff because I could see how it applied in a real situation. So I got a little more out of what those things [NOS aspects] mean, I think, than I had beforehand when we just talked about it. It was like "Oh, I get it. I see what we are doing." So I liked that. [Jay, post-interview]

Interactions with other researchers were invaluable. The interns learned about the work involved, direction of the research, impacts and assumptions inherent to the research, and NOS views of the researchers. Even interns with years of research experience claimed that the personal interactions and lab/journal club meetings were valuable for closely examining authentic research and the role of collaboration. For example, Kim reported

Departmental seminars are another example of a social activity and the members of this lab attend these seminars regularly. [Scientist mentor], in fact, gave the department seminar three weeks ago. These activities are social because there is a lot of interchange of ideas, suggestions, and encouragement between peers. I would suggest that students be required to attend lab meetings. They could see the planning, the rationale, the data, and emotions.

Brenda reported not to have gained anything from the journals or seminars. However, she did learn about the porphyrins and the chemistry research conducted in her setting:

Well for someone who has already done research before, they have already done research for years and they aren't going to gain anything. For me, the reason I gained a lot from the literature background was because the guy I was working with was recommended by the guy used to work with. I had sought him out. We kind of were coming from the same place. [Brenda, post-interview]

Brenda reaffirms her emphasis on traditional content in contrast to the nature of the research. She and Hank seemed to hold images of themselves as scientists and learners of traditional science content.

DISCUSSION

The results indicate that the science research internship, as a whole, effected desirable changes in the interns' NOS views. In general, those who made the greatest advances began the course with fairly shallow views of NOS. This result is not surprising given that they had the most to gain. Adam, Darren, Jay, Kris, Laura, and Lisa could do little more than mimic the descriptions they had heard previously. It is important to note that they also *recognized* their deficiencies during the initial interview and worked throughout the course to clarify their views. Brenda and Hank held views similar to these six, yet they held more confidence in their initial views. This finding suggests that their high efficacy impeded their advancement because they saw no need for change. The remaining interns entered the course with more developed NOS views, with Tim and George holding the most informed views. As such, they did not have as many conceptual hurdles to clear. Enhancement for these two was more in the form of supporting examples that corroborated their existing views. These results are consistent with proponents of conceptual change learning (Posner et al., 1982). Cognitive dissonance may have enabled the interns to seek new information, grapple with new ideas, and attain acceptable resolution.

Prior research background did not appear to have a differential effect on NOS learning. Kim and George had the most prior experience, and both demonstrated some enhancements. Rich and Hank held similar research backgrounds, yet only Rich showed NOS-related advancements. Brenda held little experience compared to Hank, and neither advanced. George reported having difficulty applying his NOS knowledge to his internship setting because of his limited background knowledge relative to that research area. In contrast, he had extensive knowledge of the background, assumptions, and direction of his prior research of 16 years. The task of identifying examples relevant to NOS issues was much easier for him when he considered his prior research and then attempted a comparison with his internship setting. Kim and Rich made similar claims. The more familiar context may have served as the more meaningful context for *initial* reflection. Overall, however, prior research experience did not differentially impact these interns' developments in NOS conceptions.

WHAT FACTORS OF THE SCIENCE RESEARCH INTERNSHIP COURSE INFLUENCED CONCEPTIONS OF NOS?

Active Reflection

The research experience, if considered by itself, had little impact on interns' conceptions of NOS. The lack of perceived effect on NOS views along with the reported inconsistencies in mentor scientists' conceptions of NOS lend support to the claim that just "doing science," or experiencing scientific inquiry in an authentic setting, is *insufficient* to enhance one's NOS

conceptions. There is a missing, critical factor. The results of this study suggest that explicit and guided attention to and reflection on NOS in the context of the authentic scientific research experiences is that missing factor. These findings corroborate reports promoting an explicit/reflective, constructivist-based approach to teaching about NOS through inquiry pedagogy (e.g., Bianchini & Colburn, 2000; Khishfe & Abd-El-Khalick, 2002; Roth & Lucas, 1997; Ryder, Leach, & Driver, 1999; Smith et al., 2000). The guided journal writings and seminars provided opportunities for *active* reflection. Active reflection is purposeful and explicit. These opportunities also allowed peer support and sharing. Interns reported that discussions encouraged clarification while they attempted to make NOS relevant through their experiences. The reader is reminded that most seminars were discussion-based and lead through intern–intern interactions. Based on the apparent changes in depth of NOS understandings, the guided reflections and peer sharing appeared to enable the interns to first, personalize their understandings of NOS, and second, to explore deeper issues of NOS than had been previously recognized.

Context for Reflection

Reflection requires context. The scientific inquiry context in this case was the intern's experience in the research setting *as well as* descriptions brought from others during seminar sessions. The research context enabled interns to address the question, "How does what I am experiencing, fit with my understanding of NOS?" Through these efforts to construct, define, and *refine* their own meaning of NOS, the results indicate many of their views were strengthened and deepened. Previous shaky or questionable responses turned into declarative statements accompanied by supporting examples. Deeper views were evidenced by loss of misconceptions and the recognition of interdependence among NOS aspects and broader applications. These interns only gained a "snapshot" glimpse of the research settings. They did not engage in authentic scientific inquiry *themselves*. They participated on the periphery of these *authentic contexts* through simple practices, questioning, and observation. This distinction is important. The results suggest the appropriateness of the peripheral role for enabling meaningful reflection and enhancing NOS conceptions.

A Comparison of Hank, Adam, and Brenda: Activities and Perspectives

Research settings and activities therein did not appear to have a differential impact on learning of NOS. Hank and Adam provide a good comparison because they were placed in the same setting. Their initial views of NOS were essentially the same, with Hank being only slightly more developed than Adam. Both performed the same activities, often together, for the same purposes. The level of inquiry was fairly low given that they followed strict protocols. Yet, Adam wrote extended details of his work and responses to focus questions as well as unprompted personal reflections. In contrast, Hank confined his writings to lab procedures, explanations of theories connected to the work, and minimal responses to focus questions without supporting examples. Adam demonstrated great developments in his NOS views, whereas Hank showed no change. What was the difference? There are a couple of possibilities. First, Hank's prior research experience may have led him to believe he did not have anything more to learn about research. The research was new to Adam, and he was eager for the experience. The second possibility may be related to the first. Hank's perspective of himself was as a scientist with a concentration on the content of the research, not its nature. His lack of response to focus questions compared to the elaborate reflections

written by Adam indicate they held very different perspectives of themselves in that single setting doing the same activities.

Similarly, Brenda's perspective of herself may have contributed to her lack of advancement. Relative to the other interns, she experienced a higher level of inquiry. She had a strong focus on science content, chemistry in particular, and assumed responsibility in her research setting for the design and implementation of a project. She reported thinking and acting as a "mini-graduate student." Like Hank, Brenda concentrated on the "doing of science" not the "nature of science." Like Hank, Brenda showed no growth in NOS conceptions. Comparing Hank, Adam, and Brenda suggests a third influential factor on interns' NOS developments in authentic science contexts: *Perspective*.

Perspective: Reflection from the Outside

Alice said she looked at "old experiences with a new mindset." Kim reported considering the nature of her work, past and present, with a different perspective than she ever had before. In doing so, they realized that simply doing the work of a scientist does not guarantee one will learn about NOS. This investigation suggests the perspective held by the intern is perhaps the most critical factor impacting learning of NOS in such a context. Research background, content area, research setting, and activities within the setting did not differentially affect the learning outcomes. We have shown how reflective activities and contexts for these reflections established *opportunities* for NOS enhancements. Yet, not all interns advanced. The data suggest that those participants who were more open to reflection and examining the authentic research environment through a philosophical lens made advances in their NOS learning. We are not claiming they had a true philosopher's perspective, as they had no training in a formal approach. However, the needed perspective was quite reflective in nature. Reflection is a metacognitive activity and that can serve to inform the learner of his/her developments in concept understanding and lead to deeper understanding (Hodson, 1998; Paris & Ayres, 1994). Developments in NOS conceptions required self-examination of NOS views. What does it take to become reflective within this authentic context?

The present internship model varies from a typical apprenticeship model that intends for the learner to move from the periphery to the core (Lave & Wenger, 1991) of the scientific community. The intention here is not for the learners to become members of the scientific community, but rather to become reflective *about* science within the context of the scientific community. These intentions may be subtly different in syntax, but they are vastly distinct in meaning and approach. Richmond and Kurth (1999) suggest that as learners move from the "outside to the inside" during a science apprenticeship they develop understandings about the practice and culture of science. We suggest that the ability to express and modify one's views of NOS in an authentic research context requires the learner to *shift back to the outside*, or periphery, to assume a reflective stance conducive to challenging and developing one's conceptions of NOS. Without *reflection from the outside*, one is less likely to consider the nature of the enterprise in which he/she has become a part. *Reflection from the outside* appears to require a cognitive disengagement from the authenticity of scientific activity (notably a small step for most of these interns). Maintenance of an insider's perspective maintains cognitive focus and reflection on specific content and goals of the research itself. Reports on scientists' NOS views corroborate these findings. Those most engaged in authentic scientific inquiry do not tend to, nor *need* to, explore philosophical issues surrounding their enterprise (Elby & Hammer, 2001; Glasson & Bentley, 2000). Their purpose is on process and production of the knowledge, not on epistemological underpinnings of these processes and products. Focus within one perspective does not necessarily relate to performance within the other perspective (Elby & Hammer,

2001). As Alice stated, "... They just do the inquiry, they don't philosophize about it's nature..."

Interestingly, this realization from the interns relative to their scientist mentors, was useful in reinforcing the need for explicit/reflective pedagogical attentions. The journal writings and seminar discussions served to reinforce NOS concepts that were familiar to most of these interns, but not contextualized within real science. To help their own students gain a perspective that will enable them to learn about NOS, the interns saw a need to provide experiences with scientific inquiry as a context, and they came to understand the value of explicit/reflective instruction within that context. They came to realize that meaningful learning opportunity is more than just hands-on activity. For meaningful learning there must also be purposeful teaching.

LIMITATIONS OF THE INVESTIGATION

There are several limitations to the present investigation. First, the relationships identified between factors of the internship and learning outcomes may not be generalized beyond these participants. Second, learning outcomes related to scientific inquiry and teaching of NOS were not assessed in a pre/post format. Descriptions of respective changes relied strictly on self-reports. Formal assessment of the effects of such an internship toward these learning outcomes is needed. Third, although each component of the internship served a particular role for these interns' learning about NOS, all interns experienced the same three components. As such, the requirement of individual components is not known. In particular, the requirement of the research setting is brought into question. Beyond providing a context for reflection, this component, in and of itself, did not affect a change in NOS views. Further study is needed to determine the nature of the research component as a unique necessity for enhancing learners' NOS conceptions, or if other means could be used to achieve the same ends. Other means might include scientist speakers in a seminar format, rotation through several research settings for shorter duration, web-based inquiries, and history of science studies. Their feasibility as viable authentic contexts should be explored.

The reliance on questionnaire and interview data only for the preinternship profiles, and multiple sources for postinternship triangulation, is a recognized methodological limitation to the study. The interns voiced their views throughout the internship, and these data were considered in the postanalysis. The journaling and seminars were part of the explicit/reflective instruction and, thus, the learning process. The journals and seminar activities helped the interns to "step back" from the "doing of science" to facilitate reflection "upon the science." We find that the voices of the interns in the postinternship interviews offer consistency and clear evidence for the changes reported here.

THE INSTRUCTOR'S DEVELOPING PEDAGOGICAL CONTENT KNOWLEDGE FOR NOS

Pedagogical content knowledge (PCK) for NOS is complex and interrelates multiple domains of knowledge (Schwartz & Lederman, 2002) and surfaces through multiple instructional dimensions (Bartholomew, Osborne, & Ratcliffe, 2002). One of the goals of the course was to model an explicit teaching of NOS and inquiry. The use of the journal focus questions, seminar assignments, and discussions were the mainstay of that approach. The interns prompted each other through discussion and comments during the seminars and, in a sense, were successful in using peer interactions to further their development of NOS understandings. The instructor clarified points of confusion on occasion, but it was the reflection through focus questions and class presentations that served to guide interns' thoughts in the

desired directions. In as much as the journal writings and seminars were successful toward the goals of the course, there was more that could have been done. Observations of seminar instruction and review of feedback given on journal writings revealed missed opportunities for additional guidance that would further enhance establishment of connections with inquiry experiences. A good example of missed “teachable moments” occurred during the two aforementioned episodes where Rich and Darren realized the interdependence of NOS aspects. They raised their revelations to the entire class, but the instructor did not identify the significance of their revelations. Both times she chose to move the class along to the next presenters. In hindsight, this managerial action glossed over major insights. Their “ah-hah” experiences reflected a major development in their NOS conceptions. Others in the class may have been struggling with similar thoughts but may have been unable to fully grasp the connection in the brief exchange.

Feedback in journals at times lacked necessary substance to draw interns’ attention to connections. Most often the instructor’s comments were positive, yet restricted to “ok” or “good thoughts” rather than providing meaningful guidance for interns to make deeper connections or further exploration of ideas.

An instructor’s PCK for NOS determines what and how NOS is taught (Schwartz & Lederman, 2002). This was a new course and a new teaching approach for the instructor. As such, her level of PCK for NOS developed with the course. Others have shown the importance of intentional planning and teaching of NOS (Bianchini & Colburn, 2000; Lederman et al., 2001; Schwartz & Lederman, 2002), and the role of experience and comfort in a teacher’s ability to capitalize on classroom moments through dynamic discourse (Bartholomew, Osborne, & Ratcliffe, 2002; Roth & Lucas, 1997; Schwartz & Lederman, 2002). One must be aware of what opportunities might arise, recognize when they do arise, and know how to turn those moments into successful teaching episodes. A teacher’s active reflection and experience are key elements in developing PCK for NOS.

FINAL NOTES: IMPLICATIONS FOR TEACHER EDUCATION AND TEACHING NOS

The science research internship was successful in helping strengthen and deepen these preservice teachers’ conceptions of NOS. Three factors were influential: (1) explicit opportunities for reflection through the journals and discussions, (2) the authentic context of the research setting, and (3) the reflective perspective of the intern. The results of this investigation offer support to the claim that just “doing science” is *insufficient* for one to develop conceptions of NOS as promoted in reform documents. It would seem that to *do* science, one need not understand NOS. There is a gap between inquiry and NOS. However, these preservice teachers will be teaching *about* science, not “doing” science. To *teach* science, one must have knowledge of the concepts and effective approaches to teaching science. As such, to teach NOS and inquiry, the teacher needs knowledge of NOS and inquiry, and pedagogical knowledge for each. This knowledge includes an awareness of the inherent gap between these two concepts and knowledge of how to bridge that gap. For the preservice teachers in this study, the guided discourse and writing practices were critical for NOS conceptual developments within the context of authentic research settings. If teacher educators want to effect changes in views of NOS through authentic inquiry experiences, the instructional model presented in this course is an effective approach. Moreover, the approach to teaching NOS through inquiry in the K-12 classroom should similarly involve reflective activities and explicit discussions to enable students to construct their understandings of NOS within the context of classroom-based inquiries. The results of this investigation suggest that the development of NOS views in an inquiry context can be achieved through explicit

attention to NOS issues and guided reflection, provided the learner is able to step out of the role of inquirer and assume a reflective perspective, if even for an intermittent time. Guided reflection serves as a bridge to span the gap between the inquiry context and NOS.

Success in scientific practice does not require overt consideration of science epistemology (e.g., Elby & Hammer, 2001). The purpose of teaching about NOS and scientific inquiry is to enhance scientific literacy (AAAS, 1993; NRC, 1996), not necessarily to train students to be scientists. We have suggested here that for the purpose of developing deeper conceptions of NOS, meaningful reflection requires a cognitive disengagement from the authentic context of scientific identity. This is *not* meant to suggest that scientists are incapable of meaningful reflection. This is to suggest that to consider the nature of authentic practice, one must mentally step out of that context. Thus, this is a question of transition ability. How easy is it to shift perspectives back to the outside once one has become a functioning member of the community (Wenger, 1998)? If we want learners to be reflective about scientific inquiry and NOS, we must consider their perspective of self. With this in mind, to what extent should a literal view of “students as scientists” be promoted for developing conceptions of NOS? Other quasi-inquiry contexts might serve to frame students’ thinking, but the impact from compromising authenticity presents undesirable images of NOS (e.g., Chinn & Malhotra, 2002; Driver et al., 1996). Is there a needed compromise between the extremes of authentic identity and no identity? Given the need for perceptual transition, we suggest further research to explore context dynamics, including scientific discipline, content area, investigative methods, and personal experiences within these contexts that are conducive for epistemological developments.

APPENDIX A: JOURNAL FOCUS QUESTIONS

Set #1

1. What is the context of your project? To answer this, you will need to explore the previous and ongoing research in this laboratory. In what direction is the research going? This may require you to continually ask questions about the lab’s research agenda. Clarify your project relevant to the larger picture of the research area.
2. What is your initial role in the investigation?
3. Relate the readings to your current ideas of conducting a scientific investigation. [reading was the first chapter in Science For All Americans (AAAS, 1996)]

Set #2

1. What evidence do you see of scientists using hypotheses and theories in your research placement?
2. How do scientists formulate a testable hypothesis? Do all scientific investigations require testable hypotheses? Give examples.

Set #3

1. What scientific concepts are guiding the research project that you are most familiar with in your placement?
2. Compare the research project you are most familiar with in your placement with other research experiences you have seen. How are they similar? How are they different?

Set #4

1. If there is more than one project being pursued in your laboratory, do they all follow the same general approach? If so, what is that approach? If not, how do they differ?
2. How is what you have seen in your laboratory situation different than the discussions about scientific inquiry and the NOS in the summer course? How is it the same?

Set #5

1. How are models used in the research project you are most familiar with in your placement?
2. In what ways does what is happening in your laboratory setting related to “science is a complex social activity?”

Set #6

NOS assignment.

The purpose of the assignment was to further focus their thoughts on the specific aspects of NOS emphasized in the MAT program. Through reflecting on their own experiences in their present research settings, they were to bring examples to support and/or refute these aspects. The specific examples were not most important, but rather their explanation for the presence or absence of evidence for the specified aspects was important for them to demonstrate the depth of their understanding of the NOS. The assignment included a brief presentation by groups in order to generate classroom discussion.

The students received the following assignment during the 3rd seminar and presented during the 4th.

Making Connections Between the Nature of Science and Scientific Research Experiences

Being aware of and experiencing the development of scientific knowledge, the assumptions, practices, and values that are an intricate part of the nature of science is a primary goal of the research internship course. Directly relating experiences from your laboratory setting to what you understand about the nature of science will hopefully aid in deepening your understandings beyond the surface, terminology-based views you’ve been asked to accept thus far.

For each of the seven aspects of the nature of science you’ve been introduced to, you are to look for evidence in your research setting that exemplifies or perhaps contradicts your understandings. You need to dig deeper than just providing examples, however. You need to look for and try to identify underlying assumptions, biases, and additional factors that contribute to your examples. Many examples will be appropriate for several aspects and these interrelations should be described. In other words, don’t force a separation of subjectivity, inference, and tentativeness if you can describe how these are related with one example. Your 10-minute presentation will be on the one area you previously selected.

A second part of this assignment is more reflective. You should include in your writings descriptions of where you were surprised, frustrated, and/or enlightened by your experiences. Included below are some helpful hints to get you started in the right direction. The more you are able to communicate with the people in your laboratory, the better.

1. *Tentative (subject to change)*: As each of the other six aspects of the nature of science comprise reasons leading to the tentative nature of science, this can be a really broad topic. You might consider asking the scientist in the lab or other graduate students about the models in use now, the assumptions made, and what issues are considered controversial. Information concerning the historical development of the theories, laws, and models in use in the particular research area would likely provide examples of tentativeness. On a smaller scale, you can also include instances in your laboratory where interpretations varied (because of reinterpretation or additional information) and lead to changes in the developing methods and/or models.
2. *Empirically-based*: Look for instances where conclusions were drawn and supported by multiple data sources, including the scientific literature. How are contradictions or anomalous data handled?
3. *Subjective (theory-laden)*: All decisions and conclusions made are unavoidably subjective in nature. Try to identify instances where subjectivity was openly recognized. Try to identify the assumptions (theories, other biases, etc.) that influence the decisions made in the research.
4. *Human inference, imagination, and creativity*: What phases of the investigations can you find inference? Imagination and creativity? Are these necessary factors in the progress of the research? What influences the inferences?
5. *Socially and culturally embedded*: Examine and describe the immediate social and cultural community of your laboratory setting. Do you think everyone makes the same assumptions and has the same general direction for the research (i.e. are the projects related and collaborative?)? If possible attend some journal club meetings to expand your description of the scientific community. Be observant to occasions where other research communities are discussed. Is the science in your lab considered “local” or “global?”
6. *Observations and inferences*: In addition to providing examples where observations and inferences were made, try to identify the underlying assumptions behind these, and how the assumptions influenced what was observed (data collected, interpretations of “good” or “bad” data, etc) and how the data were then interpreted.
7. *Theories and laws*: Are these terms used appropriately in your setting? What is some of the evidence in support of a theory or law driving a portion of the research? If possible, find out how the theory (or law) has changed over time.

APPENDIX B: SEMINAR SESSIONS

Introduction to the course:

Students received their placements and reviewed the course purpose and expectations. The first week’s journal questions were assigned.

In addition to debriefing research experiences and journal questions during each seminar, seminars also involved the following planned activities:

- *Seminar One: Week 3*

This seminar session was devoted mainly to group discussions about research settings and projects. Students were asked to describe their settings, the questions being investigated, and to compare settings with others in their group. Students watched a segment of a Richard Feynman in a video, “The Pleasure of Finding Things Out.”

The instructor asked the students to identify statements or viewpoints presented in the video that dealt with NOS. After some prompting, students were able to discuss several connections exemplified in the video such as subjectivity, creativity, and tentativeness of science. However, the discussion was brief and the instructor did not elaborate on student comments.

- *Seminar Two: Week 5*

Students discussed a chapter reading by Bruno Latour (1987). They were asked to find examples from the reading to support the following statements: (1) Scientific knowledge is conditional, never “proven,” in an absolute and final sense. (2) Scientific knowledge involves human imagination and creativity. (3) Scientific knowledge is subjective and theory-laden. Students discussed in three small groups and then presented to the whole class. Also during this seminar, one of the students gave a half-hour presentation on his master’s research in geology. This talk focused on the details of the research methods and results, without attention to NOS. This presentation lasted until the end of the session, and thus was not followed by class discussion on NOS aspects pertinent to the research.

- *Seminar Three: Week 7*

One of the mentor scientists gave a 30-minute talk about his research in oceanography. Questions from the students focused on the content of the research. After the presenter left, the students had a brief discussion about their impressions of the speaker’s presentation in regard to NOS and inquiry. Students engaged in an inquiry-based activity. They did the well-known “paper towel investigation.” Discussion followed connecting the activity to aspects of NOS and inquiry. No connection attempts to lab settings.

- *Seminar Four: Week 9*

Students gave 10-minute presentations of their NOS assignments (Appendix A, set #6). Class discussions followed each presentation. Students and instructor asked questions and commented on specific details of the interns’ research experiences, specific aspects of NOS, and connected aspects of NOS. The issue of variety of approaches in scientific inquiry was also raised during the session.

- *Seminar Five: Week 10*

One of the mentor scientists gave a 45-minute talk about his research on mass spectrometry and his views of science teaching, followed by a question and answer session with the students. The instructor introduced an Inquiry Scoring Guide and discussed the limitations of the rubric.

APPENDIX C: POSTINTERNSHIP INTERVIEW

Phase One: Research setting and experiences

1. Please describe your research setting and the research experiences you had within your setting.
2. Were you involved in any specific research projects? If so, what types and what did you do?
3. [If involved in an investigation:] To what extent was your investigation guided by others within the research setting? How much input did you have into the design and implementation of the investigation?
4. How would you describe the general atmosphere of the research setting?
5. How would you describe the lead scientist? What types of activities was he/she involved in?

6. Did you become familiar with the other people in your research setting? If so, what type of work were they doing?
7. [If participated in laboratory meetings or journal clubs:] What type of discussions took place during these times?
8. Were you able to actively participate in these meetings?
9. What was the purpose of the meetings?

Phase Two: Perceived learning outcomes and attributions for learning outcomes

1. In your opinion, what, if anything, did you learn or gain from the research internship?
2. Why (or why not) do you feel you benefited in that way?
3. Provide examples, if possible, of how the internship has affected you.

Phase Three: Perceived changes in views of the nature of science and corresponding attributions

Participants are provided with their postinternship nature of science questionnaire and asked to elaborate and explain responses as in the first interview. They are also provided with their preinternship questionnaire and asked to compare their responses. Participants are asked:

1. Have your views or level of understanding of the nature of science changed in any way from your views at the start of the internship? If so, how?
2. How, if at all, has the internship influenced your views on the nature of science?
3. [If response to #2 is negative, yet views have changed:] To what do you attribute the change in your views or level of understanding?
4. [If response to #1 and #2 are negative:] Why do you think your views or level of understanding of the nature of science has been stable?
5. Consider the research experience, the journals, and the seminar sessions. Do you think any of these components of the internship influenced your views of the nature of science? If so, what components? How? And Why?
6. Can you recall examples or specific instances that you feel had an influence on your understanding? Explain.

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